ModelIt!

Teacher's Guide

Hand Warmers and Hidden Reactions

How Atoms Rearrange to Create Heat, Light, and Everything You Use

Quick Reference

Grade Level	Grade 8
NGSS Standard	MS-PS1-4, MS-PS1-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS1-4, MS-PS1-5

Develop a model that predicts and describes changes in particle motion, temperature, and state of a pure substance when thermal energy is added or removed. Develop and use a model to describe how the total number of atoms does not change in a chemical reaction and thus mass is conserved.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model to describe how atoms rearrange in a chemical reaction while total mass is conserved.
Disciplinary Core Idea	**PS1.B Chemical Reactions** — Substances react chemically to form new substances with different properties. The total number of atoms is conserved.
Crosscutting Concept	**Energy and Matter** — Students trace how matter (atoms) is conserved while energy is released as thermal energy in an exothermic reaction.

CAST Testing Alignment

MULTIPLE CHOICE:

Iron powder (10g) reacts with oxygen from the air inside a sealed hand warmer. After the reaction is complete, what is the total mass of the products?

CONSTRUCTED RESPONSE:

Using the law of conservation of mass, explain what happens to the atoms during the chemical reaction inside a hand warmer. Where does the heat come from if atoms are neither created nor destroyed?

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Hand warmers are elegant examples of chemistry in action. The simple exothermic reaction of iron oxidation demonstrates both conservation of mass and energy transformation — making abstract chemistry concepts tangible and practical.

The Chemistry of Hand Warmers

Hand warmers contain: iron powder (reactant), salt (catalyst to speed the reaction), activated carbon (distributes heat and retains moisture), and vermiculite (insulation). When exposed to air, the iron reacts with oxygen: $4Fe + 3O_2 \rightarrow 2Fe_2O_3$ + heat. This is the same reaction as rusting, but engineered to happen faster.

Exothermic vs. Endothermic

Exothermic reactions release energy to the surroundings (hand warmers, combustion, cellular respiration). Endothermic reactions absorb energy from the surroundings (instant cold packs, photosynthesis, cooking an egg). The direction of energy flow determines whether you feel heat or cold.

Conservation of Mass

Antoine Lavoisier discovered that mass is conserved in chemical reactions. In a hand warmer: the mass of iron + the mass of oxygen that reacts = the mass of iron oxide produced. No atoms are created or destroyed — they simply rearrange. If you weigh a sealed hand warmer before and after, the mass is identical.

Reaction Rates

Several factors affect how fast a reaction occurs: surface area (powder reacts faster than a solid chunk), temperature (higher temp = faster), concentration of reactants (more oxygen = faster), and catalysts (salt speeds up the iron oxidation). Hand warmers are engineered to balance these factors for consistent, safe heat.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify iron powder amount, oxygen exposure, reaction rate, and heat output as system components.

E — Establish Relationships
Students determine that both iron and oxygen positively affect reaction rate, and reaction rate positively affects heat output.

V — Visualize & Model
Students build the chemical reaction model showing how inputs drive the reaction and produce heat.

E — Evaluate Outcomes
Students compare sealed (no reaction), normal use, and maximum exposure scenarios.

R — Revise & Extend
Students add salt (catalyst) or insulation to model real hand warmer engineering.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"This little packet creates heat for 8 hours. No batteries. No fire. No electricity. How?"*

Watch For

Give students hand warmers to feel. Where does the heat come from?

Slide 2: Learning Objectives

3 min

Say: *"Today we discover how atoms rearranging can create heat energy."*

Watch For

Define exothermic with everyday examples (warm things = exothermic).

Slide 3: Big Question

3 min

Say: *"No fuel is burning. No electricity flows. But it's warm. Chemistry is happening!"*

Watch For

Quick write: Where do you think the heat comes from?

Slide 4: LEVER Framework

2 min

Say: *"We'll model the chemical reaction that turns iron and air into heat."*

Watch For

Preview the LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What ingredients does this reaction need? What does it produce?"*

Watch For

Open a hand warmer. Identify the iron powder. Discuss what happens when exposed to air.

Slide 6: Activity 2: Connections

8 min

Say: *"Both iron AND oxygen are needed. Take away either one and the reaction stops."*

Watch For

Demonstrate: seal a hand warmer in a plastic bag — it stops getting warmer.

Slide 7: Activity 3: Simulation

10 min

Say: *"What happens if we give the iron MORE oxygen? Less oxygen? No oxygen?"*

Watch For

Students predict, then run. Focus on the limiting reactant concept.

Slide 8: Discoveries

5 min

Say: *"The atoms didn't disappear — they rearranged. Iron + oxygen = iron oxide + HEAT."*

Watch For

Weigh a sealed hand warmer before and after. Same mass! Conservation proven.

Slide 9: STEM Challenge

12 min

Say: *"A hiking company hired you to design a better hand warmer. Use chemistry!"*

Watch For

Teams design hand warmers with specific heat and duration targets.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Iron Powder Amount, Oxygen Exposure
- **INTERNAL:** Reaction Rate, Heat Output

Relationships

- **Iron Powder Amount to Reaction Rate:** POSITIVE (+) — More iron powder means more reactant available, allowing a faster and longer-lasting reaction.
- **Oxygen Exposure to Reaction Rate:** POSITIVE (+) — More oxygen allows more iron atoms to react simultaneously, speeding up the reaction.
- **Reaction Rate to Heat Output:** POSITIVE (+) — A faster reaction releases more heat energy per unit time, making the hand warmer hotter.

Key Discoveries

- Iron + Oxygen → Iron Oxide (rust) — this reaction RELEASES heat
- The same atoms exist before and after — they just rearranged (conservation of mass!)
- More reactants AND more oxygen = faster reaction = more heat
- The hand warmer eventually stops because the iron runs out — a limiting reactant!

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Chemical reactions destroy matter"	No atoms are created or destroyed in a chemical reaction — they only rearrange. The total mass of reactants ALWAYS equals the total mass of products. This is the law of conservation of mass.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Heat means something is burning"	Exothermic reactions release heat but don't require visible combustion. Rusting, dissolving, and neutralization all release heat without flames. 'Burning' is just one type of exothermic reaction (combustion).
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The hand warmer creates energy"	The hand warmer doesn't CREATE energy — it TRANSFORMS chemical potential energy stored in the bonds of iron and oxygen into thermal energy. Energy is conserved, just like mass.